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Contraception 81 (2010) 215–222

Contraception

Original research article

Bleeding patterns and menstrual-related symptoms with the continuous use of a contraceptive combination of ethinylestradiol and drospirenone: a randomized study

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Received 9 June 2009; revised 9 September 2009; accepted 27 October 2009

Abstract

Background: The objective of this study was to compare bleeding patterns of women using a contraceptive combination of 30 mcg ethinylestradiol (EE) and 3 mg drospirenone (DRSP) continuously or cyclically. Menstrual-related symptoms were also evaluated.
Study Design: This open, prospective, randomized study evaluated 78 women using the EE/DRSP combination continuously for 168 days or for six 28-day cycles, each followed by a 7-day hormone-free interval. A diary with pre-established scales was used to assess the frequency and intensity of bleeding and menstrual-related symptoms.
Results: Amenorrhea increased with continuous use; 62.2% of women with continuous use were amenorrheic at the end of treatment (95% CI: 46.6–77.8%). Dysmenorrhea, headache, acne, nausea, edema and increased appetite improved significantly in the continuous-use group, and mastalgia and edema in the cyclic-use group ($p<.05$). Six subjects in the continuous-use group (15.4%) and three in the cyclic-use group (7.7%) discontinued due to adverse events.
Conclusions: Continuous use was associated with amenorrhea and fewer menstrual-related symptoms compared to cyclic use.
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Keywords: Oral contraceptive; Ethinylestradiol; Drospirenone; Continuous use; Bleeding patterns

1. Introduction

The conventional use of combined oral contraceptives (COCs) generally consists of cyclic administration for 21 days followed by a 7-day hormone-free interval during which bleeding occurs as a result of hormone deprivation. Currently, the benefits of this monthly pause in the use of OCs have been questioned, since, from a biological point of view, the artificial bleeding resulting from hormone deprivation does not appear necessary and may be associated with menstrual-related syndromes [1–3]. For some women, monthly bleeding is undesirable, not only because of the symptoms that appear around this time but also for personal reasons such as convenience and practicality [4]. Consequently, alternative regimens have been introduced, including continuous contraception or regimens in which the number of active pill days is extended. In both regimens, the objective is to avoid problems related to the monthly pause in contraceptive use and to improve compliance with the method, thereby increasing satisfaction and enhancing the quality of life of women who choose to use COC [5–8].

The majority of studies evaluating the continuous use of COCs have focused on bleeding patterns [9,10], since most women opting for this contraceptive regimen do so because they wish to eliminate their bleeding episodes. Others, however, extend COC use in an attempt to improve symptomatology experienced during the pause in OC use [5].

The effects of modern contraceptives in which low doses of estrogen are combined with selective progestogens have

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doi:10.1016/j.contraception.2009.10.010

also been studied in extended regimens. Similarly, the antiandrogenic and antimineralocorticoid properties of drospirenone have been evaluated, with results showing that amenorrhea rates of over 80% in continuous-use regimens are associated with a reduction in symptoms such as mastalgia, edema, swelling, seborrhea and dysmenorrhea [11,12]. Other studies carried out recently with the contraceptive combination of 3 mg of drospirenone (DRSP) and 30 mcg of ethinylestradiol (EE) also showed that the extended use of this COC for up to 1 year effectively controlled menstrual-related symptoms [13,14]. It should be emphasized that the abovementioned studies evaluated women using the EE/DRSP combination continuously in nonrandomized studies [12–14]. Due to the lack of comparative, randomized studies involving the EE/DRSP combination in a continuous regimen, the proposal of the present study was to compare bleeding patterns in users of a COC containing EE and DRSP either in a continuous regimen or during cyclic use with a 7-day pause after 28 days of use and to assess the effects of these two regimens on menstrual-related symptoms.

2. Materials and methods

2.1. Study design

An open-label, prospective, randomized clinical trial was carried out in 78 women randomly allocated to use a COC containing EE (30 mcg) and DRSP (3 mg) either continuously (continuous-use group) or with a 7-day pause (cyclic-use group).

Continuous use consisted in the daily administration of the contraceptive pill uninterruptedly for 168 days. Cyclic use consisted in the daily administration of the same contraceptive pill for six 28-day cycles with a 7-day hormone-free interval at the end of each cycle. The subjects were randomized into two groups of equal size (39 subjects in each group) to receive the combination of 30 mcg EE and 3 mg DRSP either continuously (continuous-use group) or cyclically (cyclic-use group). For the randomization process, a computer-generated list was created using the Random Allocation software program, version 1.0, May 2004, originally designed by Dr. M. Saghali, University of Medical Sciences, and freely available to the public. The subjects were allocated sequentially according to the pre-established randomization number, using sequentially numbered opaque envelopes which contained the randomization number. The randomization process and the statistic analysis were done by an independent specialist unrelated to the study.

The protocol was approved prior to study initiation by the internal review boards of each participating center, and the study was conducted in accordance with the directives of the Brazilian regulatory authorities (ANVISA), with the guidelines of Good Clinical Practice established by the International Conference on Harmonization (ICH) and with the principles defined in the Declaration of Helsinki. All women were informed with respect to the nature of the study and voluntarily agreed to participate by signing an informed consent form.

2.2. Participants

Women at three Brazilian health care centers, who were sexually active, requesting a contraceptive method and not currently using any form of hormonal contraception or an IUD, were eligible to participate in the study. The inclusion criteria consisted of age 18–35 years, body mass index (BMI) of 19–30 kg/m^2, a minimum of 8 years of schooling, ability to understand oral and written instructions, and willingness to comply with the study requirements and to sign the informed consent form. Women were not admitted to the study if they were pregnant (either suspected or confirmed pregnancy), if they had an abnormal Pap smear, a personal history of genital or breast neoplasia, cardiovascular disease, arterial hypertension, venous or arterial thromboembolism, diabetes mellitus, acute or chronic hepatopathies, or alcoholism. In addition, women who had been using hormonal contraceptives in oral, injectable or implant form in the 2 months preceding the study and those in concomitant use of barbiturates, hydantoins, carbamazepine, phenylbutazone, meprobamate, rifampicin or any other drugs capable of interacting with OCs were excluded from the trial.

The woman's medical history was recorded and a complete physical and gynecological exam was carried out, including sampling for cervicovaginal cytology. The following laboratory tests were requested as safety parameters: complete blood count, fasting glucose, AST, ALT, total and fractionated bilirubin, gamma-glutamyl transferase, urea, creatinine, alkaline phosphatase, TSH and free T4 and a beta-hCG pregnancy test.

Two weeks later, if the test results were satisfactory and the inclusion and exclusion criteria were met, the women were randomized and instructed to initiate use of the COC containing EE (30 mcg) and DRSP (3 mg) on the first day of their next menstrual period and to use it either continuously or with a 7-day interval without hormone use at the end of each 28-day period. The study drug was supplied in packs of 28 pills and each patient received a total of six pill packs.

Each woman was requested to complete an individual diary specifically designed to record information on the date of initiation of pill use, the number of forgotten pills, the occurrence of bleeding and its intensity, and any symptoms experienced during the evaluation period.

The women who were randomized to continuous treatment used the contraceptive pill for 24 consecutive weeks, returning to the clinic every 4 weeks for a follow-up visit. In the group of women using the pill cyclically, the pills were taken in six consecutive 28-day cycles with a 7-day hormone-free interval between each cycle. These women returned for follow-up visits every 5 weeks. At all visits, information on bleeding and symptoms recorded in the woman's diary was evaluated, any questions were answered and any complaints or adverse events were recorded.

Clinical and laboratory exams were repeated at the end of the study period and a final follow-up visit was made.

2.3. Evaluation of bleeding and menstrual-related symptoms

In order to analyze the different parameters evaluated in this study, a diary specifically designed for this purpose was used by all participants of both groups to record data referring to bleeding and symptoms associated with the use of the contraceptive pill. For this purpose, spotting was defined as slight bleeding for which sanitary protection was not required, while mild, moderate and heavy bleeding was defined as any bleeding that did not conform to the woman's usual bleeding pattern, using the woman's previous menstrual cycles as a parameter for comparison. Bleeding intensity was scored as follows: 0=no bleeding; 1=spotting; 2=slight bleeding; 3=moderate bleeding; and 4=heavy bleeding. Menstrual-related symptoms were evaluated in the two treatment groups according to the occurrence of the principal symptoms known to be associated with cyclic use of the contraceptive pill (acne, dysmenorrhea, headache, mastalgia, nausea, irritability, edema, increased appetite, breast swelling). The symptoms were evaluated in accordance with a score reflecting their intensity (0=absent; 1=mild; 2=moderate; and 3=intense). With respect to both bleeding and menstrual-related symptoms, the baseline score referred to spontaneous menstruation immediately prior to contraceptive use in both groups.

2.4. Statistical analysis

2.4.1. Sample size and statistical power

The primary objective of this study was to evaluate the absence of bleeding induced by the COC. Based on the relevant literature [12,13], a rate of amenorrhea of 80% was expected after 6 months in women using the COC continuously and 1.6% in those using the pill cyclically. Therefore, a sample of 16 evaluable research subjects would be sufficient to achieve a statistical power of 80% and a significance level of 5%. A dropout rate of 35–40% was estimated; therefore, the analysis of the two-sided hypothesis by the two-group chi-square test established a sample size of 24 subjects, 12 for each treatment group, with 5% significance and 80% power.

We opted to include 78 subjects in order to study and try to define the symptoms improvement rate as secondary parameters. The menstrual-related symptoms were described secondarily and were not taken into consideration in calculating sample size.

2.4.2. Statistical methods

Intention-to-treat (ITT) analysis was carried out including all the women who had been randomized to one of the two treatment groups. In the ITT analysis, the final scores recorded for those subjects who discontinued or dropped out of the study were repeated for the missing visits. In addition, per protocol analyses, restricted to those women who completed 6 months of treatment or who stopped treatment due to method failure, concomitant disease or adverse events, were performed.

Qualitative variables were expressed as absolute (*n*) and relative (%) frequencies. Quantitative frequencies were described as means, standard deviations, medians and ranges. Analyses based on parametric tests (ANOVA) and nonparametric tests (Mann–Whitney and Friedman) were performed depending on the distribution found. Analysis of the qualitative variables was based on nonparametric tests (chi-square or Fisher's Exact Test).

Analysis of amenorrhea rates was performed using the chi-square test or Fisher's Exact Test to evaluate differences between the two groups.

For the analysis of menstrual-related symptoms, each symptom was calculated as a percentage of the women in whom each symptom occurred and the two regimens were compared using McNemar's test.

The results were analyzed using the Statistica software program, version 5.0 (StatSoft, Tulsa, OK, USA), and the significance level adopted was .05 (α=5%).

3. Results

One hundred and eight women were screened, 78 of whom were randomized to one of the two study groups (39 to the continuous-use group and 39 to the group using the pills cyclically). One woman randomized to the cyclic-use group did not wish to continue in the study and was therefore excluded prior to initiation of the medication. Consequently, the ITT analysis was performed on the data obtained from the 77 women who initiated treatment (39 women in the continuous-use group and 38 in the cyclic-use group).

During the study period, 10 subjects in the continuous-use group (25.6%) and nine subjects in the cyclic-use group (23%) were discontinued for several reasons. There was no statistically significant difference between the two groups with respect to study discontinuation rates (p=.8422). Fig. 1 shows the participant's flowchart throughout the study, including detailed information about the discontinuation rates and number of subjects for the final analysis.

Overall, 29 women (74.4%) completed the study in each of the two treatment groups.

The two groups were homogeneous with respect to baseline clinical parameters as shown in Table 1.

The amenorrhea rates in both groups are summarized in Fig. 2. The percentage of women reporting no bleeding episodes was significantly higher in the continuous-use group compared to the cyclic-use group in all the periods evaluated except during use of the first pack of pills. During use of the sixth pack of pills, 62.2% (95% CI: 46.6–77.8%) of the women in the continuous-use group were amenorrheic compared to only 17.7% (95% CI: 4.9–30.5%) in the cyclic-use group. Spotting, on the other hand, occurred more frequently in women in the continuous-use group from the third pill pack onwards; however, no statistically significant

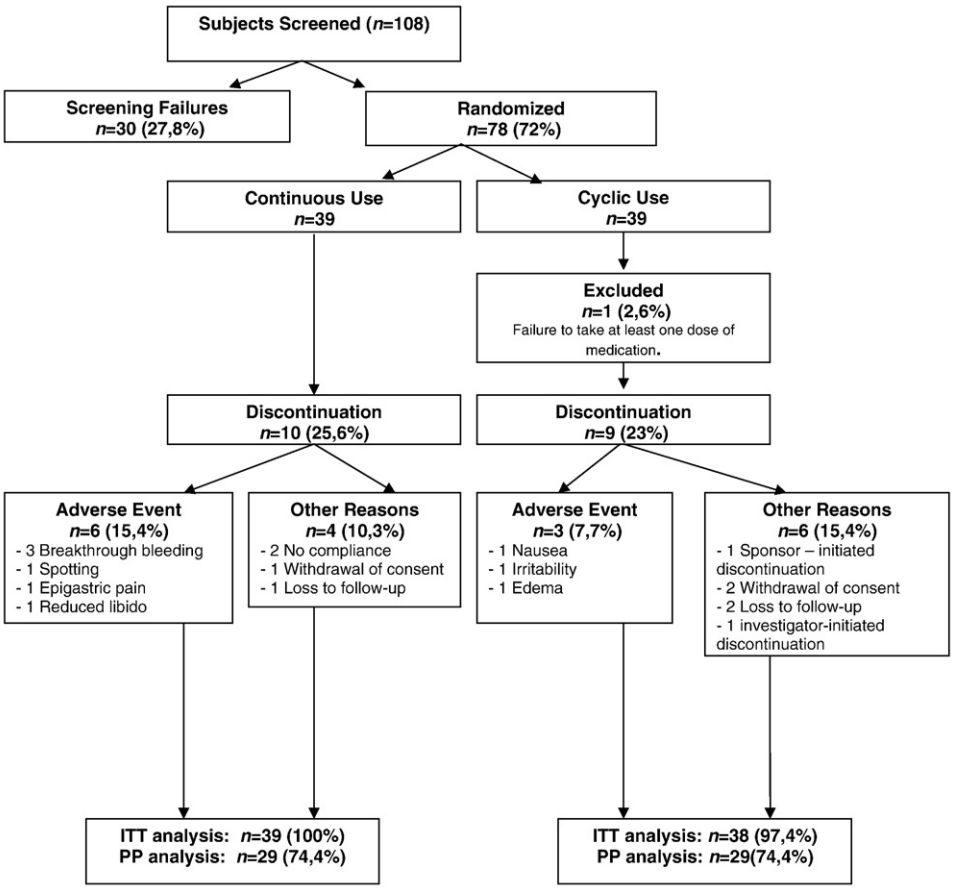


Fig. 1. Participant's flowchart throughout the study.

difference was found between the groups when the entire observation period was considered. During use of the sixth pill pack, 16.2% of women in the continuous-use group and 8.8% of women in the cyclic-use group experienced spotting (Fig. 3). The median scores for the reported intensity of bleeding were also significantly lower in the continuous-use group compared to the cyclic-use group in Pill Packs 2, 3 and 6 (Fig. 4).

Table 1
Clinical characteristics of women at enrollment

Characteristic	Continuous-use group	Cyclic-use group	p value
Age (years)	27.9±4.4*	27.7±5.1*	.8444
Age at menarche (years)	12.5±1.2*	13.1±1.3*	.0683
Length of menstrual cycle (days)	29.1±2.8*	29.3±2.4*	.6572
Smokers, *n* (%)	4 (10.3%)	4 (10.6%)	1.00
Previous use of COCs, *n* (%)	6 (15.4%)	6 (15.8%)	1.00
BMI (kg/m^2)	22.5±2.9*	22.3±2.9*	.7969
Systolic blood pressure (mmHg)	109.7±10.4*	112.1±9.1*	.2913
Diastolic blood pressure (mmHg)	72.8±6.9*	71.8±7.7*	.5565

* Standard deviation of the mean.

There was a decrease in the frequency of symptoms such as dysmenorrhea, headache, nausea, acne and increased appetite at the end of the study period in both groups; however, this was only significant in the group of women using the contraceptive combination continuously (Fig. 5). The frequency of edema decreased significantly in both groups; however, mastalgia decreased significantly only in the cyclic-use group. Comparing the use of the first pack of pills with the sixth pack, the present study found no statistically significant difference between the two periods

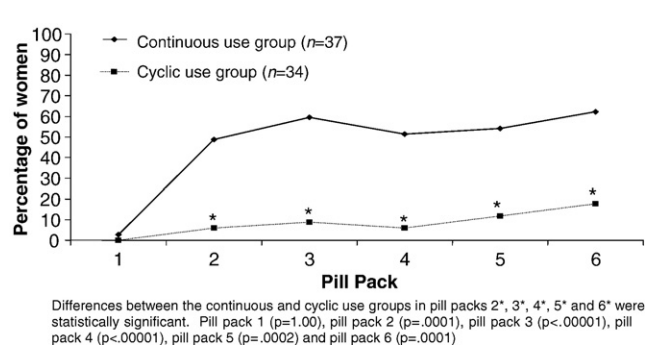


Fig. 2. Percentage of women with amenorrhea in each pill pack period of continuous- or cyclic use of an oral contraceptive pill.

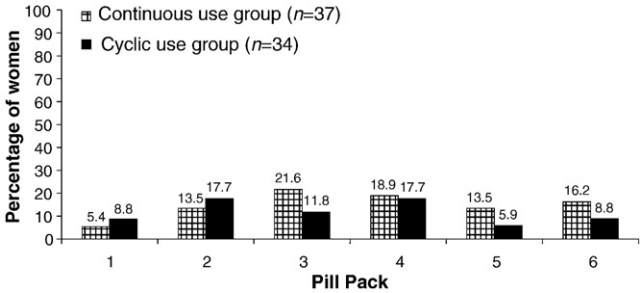


Fig. 3. Percentage of women with spotting in the two treatment groups.

of use with respect to the number of women reporting irritability or breast swelling in either of the two treatment groups (Fig. 6). Other variables such as body weight and arterial pressure remained stable in both groups throughout the observation period.

4. Discussion

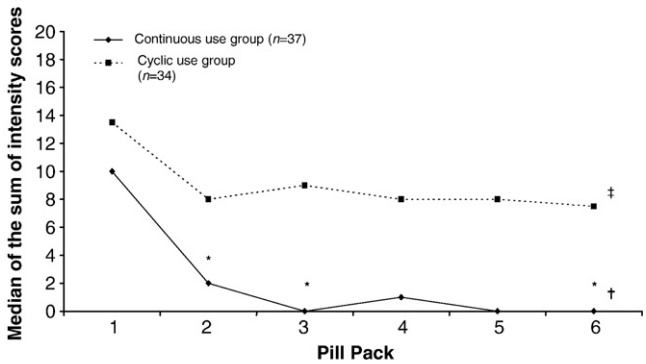
The present study shows that amenorrhea rates were significantly different between the two regimens evaluated. With respect to the continuous-use regimen, the amenorrhea rate was found to increase with the duration of contraceptive use, reaching 62.2% at the end of the study. This finding is in agreement with previous reports that showed an increase in the rate of amenorrhea as a function of the duration of treatment in women using oral contraceptives continuously [3,5,7]. This rate of amenorrhea is less than that reported in a study carried out in Germany using the same contraceptive formulation in which 85% of women became amenorrheic after 126 days of treatment [12]. Similar amenorrhea rates have also been observed in women using other oral contraceptive formulations continuously [9,15,16]. In one study, the continuous use of a contraceptive pill containing 20 mcg of EE and 90 mcg of levonorgestrel resulted in an amenorrhea rate of 66.2% by the sixth pill pack, which is similar to the rates reported in the present paper.

However, one important weakness of this and the previous reports is that it is not possible to infer individually whether the subject who was amenorrheic at the end of the study was also amenorrheic throughout the whole trial. Even though this brings important limitations on the individual assessment of bleeding patterns, it provides information on the amenorrhea rates of the studied population as a whole during each specific period of the trial.

Although the amenorrhea rates in the continuous DRSP regimen group are similar to those reported in the literature, the percentage of women reporting amenorrhea in the cyclic-use group was 17.7% at the end of treatment. This percentage is much higher than that reported with the traditional use of contraceptive pills, i.e., in a 21- or 24-day schedule followed by a 7- or 4-day pause, respectively [2,6]. In parallel, a low rate of irregular bleeding was observed in women using the pills cyclically. These findings may be a consequence of the extension of the active period of the pill to 28 days while retaining a 7-day hormone-free interval, which may enhance its effects on both gonadotrophin and endometrial suppression. It is also possible that prolongation of the hormone-active interval while retaining the same hormone-free interval may also have a more favorable impact on withdrawal symptoms. However, this is only speculative at the present time since there are no published studies comparing the 21/7 and 28/7 regimens in relation to bleeding patterns and hormone withdrawal symptoms.

It is important to emphasize that most of the bleeding reported by women in the continuous-use group consisted of spotting or light bleeding, which was not intense enough to affect their compliance or the discontinuation rates, an issue of the utmost practical importance. The spotting rate in the continuous-use group was also comparable to that reported in other studies that evaluated the use of oral contraceptives in extended regimens [16,17]. In three studies that evaluated the occurrence of spotting in women using a combination of 30 mcg EE and 3 mg DRSP in the conventional cyclic regimen, the reported rates were comparable to those found in the present study [18–20]. In a systematic review, Edelman et al. [2] found that in the majority of studies the occurrence of spotting was similar in the two therapeutic regimens.

In the present study, the median of the sum of the intensity scores for bleeding revealed that both cyclic and continuous regimens caused a significant reduction in this parameter throughout treatment. Nevertheless, there were statistically significant differences in the intensity of the various forms of bleeding (spotting, light, moderate or heavy bleeding) between the two groups during use of the second, third and sixth pill packs at which time median intensity was significantly lower in the continuous regimen, thus promoting a high rate of compliance with the method and rendering benefits to those women who wished to or needed to reduce



Bleeding intensity scores: 0 = no bleeding; 1 = spotting; 2 = slight bleeding; 3 = moderate bleeding; and 4 = heavy bleeding. Defined as any bleeding that did not conform to the woman's usual bleeding pattern, using the woman's previous menstrual cycles as a parameter for comparison.

Statistically significant differences between the continuous and cyclic groups in pill packs 2, 3 and 6 (* p<.05), (Mann-Whitney test).
Statistically significant differences between pill packs 1 and 6 in the continuous and cyclic groups (†p<.00005; ‡p<.00001, respectively), (Friedman's test).

Fig. 4. Bleeding intensity throughout the six-pill-pack period of treatment in the continuous- and cyclic-use groups.

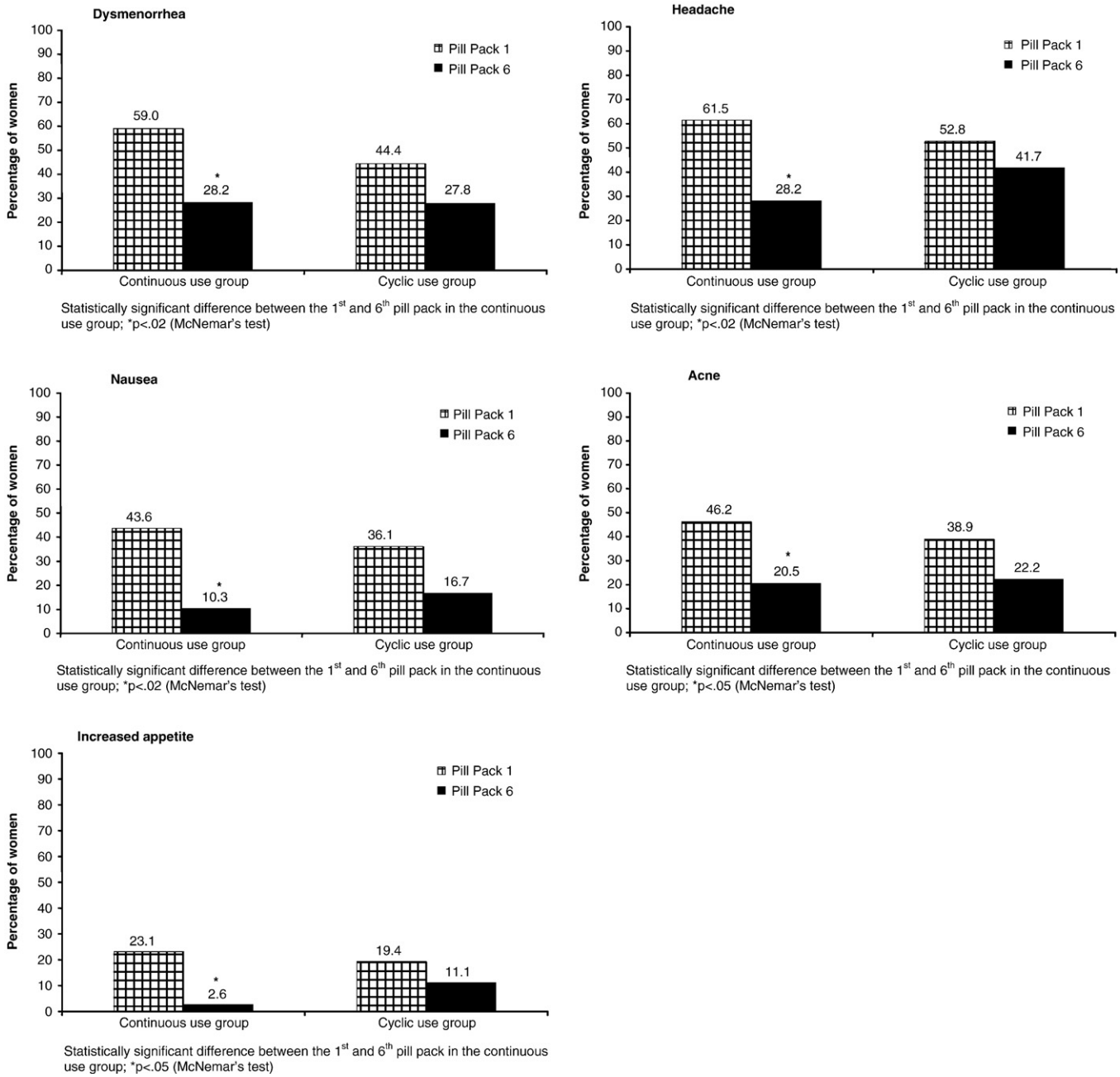


Fig. 5. Frequency of symptoms of dysmenorrhea, headache, nausea, acne and increased appetite at Pill Packs 1 and 6.

the frequency or intensity of their menstrual bleeding. These results are also in agreement with those of Miller and Hughes [9] and Kwiecien et al. [10], who compared bleeding patterns in users of the contraceptive combination of 20 mcg of EE and 100 mcg of levonorgestrel in a continuous regimen vs. a cyclic regimen for 336 and 168 days, respectively. It should be emphasized that both studies were randomized, although the present trial is the only randomized study using this particular contraceptive formulation (EE/DRSP).

The frequency of reported symptoms such as dysmenorrhea, headache, nausea, acne and increased appetite decreased by the end of treatment in both groups, although this reduction was statistically significant only in the group of women using the EE/DRSP combination continuously. These results are in agreement with previous studies showing that menstrual-related symptoms respond better to treatment when oral contraceptives are used continuously [2,5,6,11,14]. Because continuous contraceptive regimens are more effective for the treatment of menstrual symptoms than cyclic regimens, the reduction in these symptoms was statistically significant despite the small sample size.

On the other hand, the incidence of mastalgia only decreased significantly in the group of women using EE/DRSP cyclically. This finding may be due to the greater

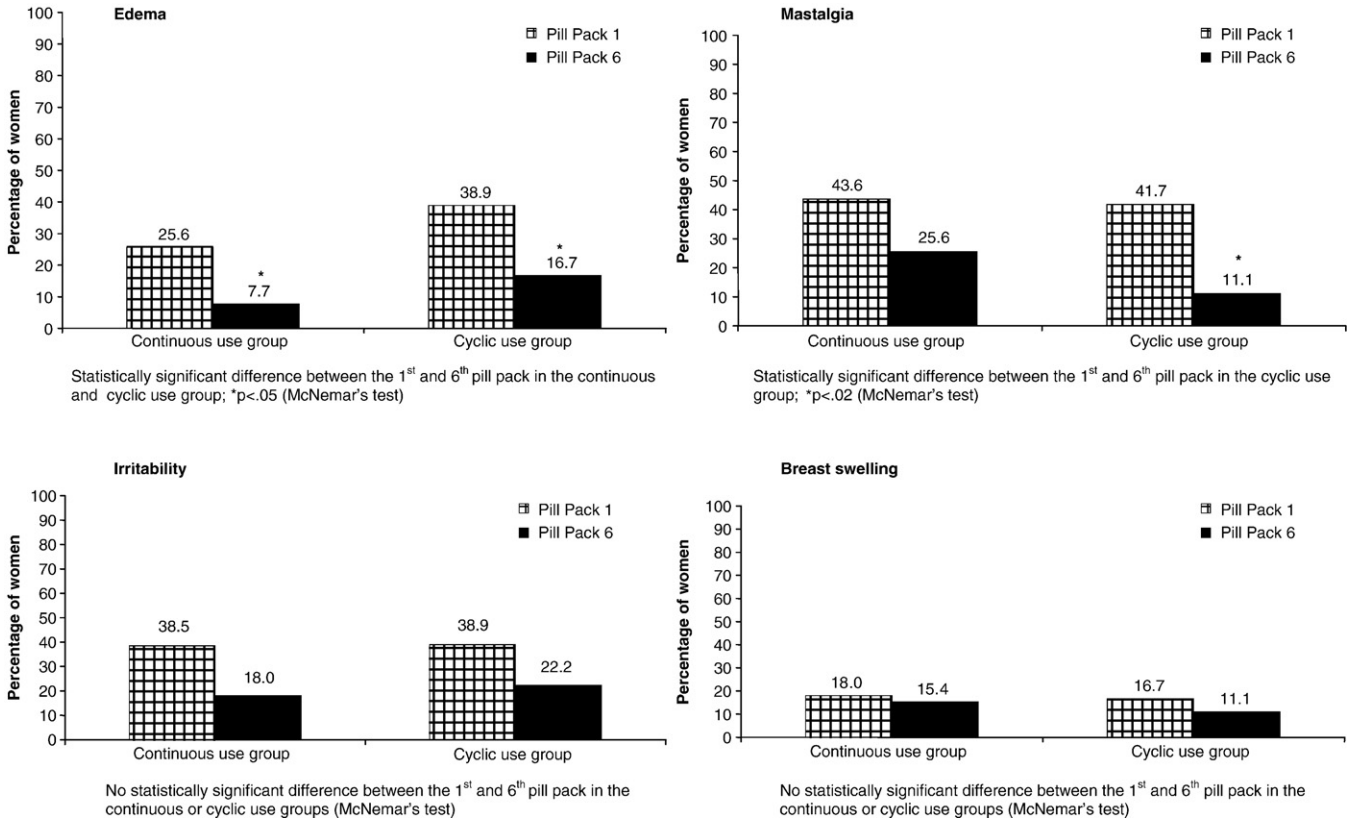


Fig. 6. Frequency of symptoms of edema, mastalgia, irritability and breast swelling in the first and sixth pill packs of treatment.

number of days of exposure to estrogens in the continuous-use regimen. With respect to breast swelling and the frequency of reported irritability, no difference was found between the two treatment regimens.

However, it is necessary to emphasize that, in our study, the statistically significant differences found with respect to the menstrual-related symptoms refer exclusively to this sample and that these results cannot be extrapolated in view of the lack of statistical representation of the sample.

A previous study that compared the use of the same hormone combination containing DRSP in a continuous or cyclic regimen showed that the improvement in symptoms such as dysmenorrhea, edema and breast tenderness was more marked in the continuous-use group [12]. In that study, a 21-day cyclic regimen with a 7-day hormone-free interval was used instead of the 28/7 regimen used in the present study; therefore, it is reasonable to suppose that with the exception of mastalgia the similarities in the results between the two studies may be a consequence of the elimination of the hormone-free interval in the continuous-use regimen. Although the 28-day cyclic regimen with a greater number of active pills may exert a greater effect on the abovementioned symptoms, the presence of the hormone-free interval still continues to constitute a limiting factor to the ability of the contraceptive pill to reduce menstrual-related symptoms in comparison with regimes in which the number of days of active pill use is extended [1,2]. The absence of variations in blood pressure and the low incidence of side effects reported in the present study are also in agreement with a previous study comparing the two regimens of DRSP [12].

In conclusion, the contraceptive combination of EE and DRSP, when used in a continuous regimen for 168 days, was capable of inducing amenorrhea and of reducing the number of bleeding days more effectively than the cyclic regimen. The former schedule also appears to be more effective than the latter for the control of menstrual-related symptoms such as dysmenorrhea, headache, nausea, increased appetite and acne.

Acknowledgment

The authors are grateful to Libbs Farmaceutica Ltda., São Paulo, Brazil, for the financial support of this study.

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